

Signed residual dynamics and a local noisy crossing certificate

Research continuation; initialization-to-entry remains open

Abstract

The repository’s strong-block statistic is the derivative of a smooth parameter functional. Its signed residual representation separates strong- and weak-cluster forcing without discarding complement neurons or training gate changes. We prove local comparison and enclosure lemmas and give a computer-assisted, fixed-dataset reversal certificate from a small neighborhood of a learned canonical state. The full OOD error decreases and rebounds on an explicit sector; weak-cluster forcing dominates at the late endpoint. The certificate is conditional on reaching its entry neighborhood. The canonical initialized reachability inequality and the high-probability iid theorem remain open.

1 Conventions and the signed weak source

Let $f_\theta(x) = \sum_{i=1}^h w_i(u_i^T x)_+$ in $\mathbb{R}^2$, $e(x) = x - f_\theta(x)$, and $\mathcal{L} = \frac{1}{2}\mathbb{E}_n\|e\|^2$. Use the Euclidean parameter norm and the empirical output inner product $\langle v, z \rangle_n = \mathbb{E}_n v(x)^T z(x)$. Throughout, the flow is compatible gradient flow with the ReLU chain rule and loss dissipation; all differential identities are almost everywhere. Training selectors are the actual selectors. The cohort I is fixed, including when it was chosen by an offline ID rule. Set

$$M = \sum_{i \in I} w_i u_i^T, \quad v = (r, -1)^T, \quad Y = e_1^T M v - r, \quad \ell = (Y, 1)^T, \quad 0 < r < 1.$$

The repository statistic, without inserting geometric k factors, is

$$Q_G = Y(r\dot{m}_{11} - \dot{m}_{12}) + r\dot{m}_{21} - \dot{m}_{22} = \ell^T \dot{M} v. \quad (1)$$

Here v is an auxiliary vector, not the unit OOD probe $\xi = (r, -\sqrt{1-r^2})$.

Write $u_i = (a_i, b_i)$ and $w_i = (c_i, d_i)$. On the weak cluster, define

$$\begin{aligned} F_I(x) &= \sum_{i \in I} c_i(u_i^T x)_+, & F_O(x) &= f_1(x) - F_I(x), \\ H_I(x) &= x_2 \sum_{i \in I} \alpha_i(x) c_i d_i, & L_I(x) &= x_2 \sum_{i \in I} \alpha_i(x) c_i (d_i - b_i), \\ Z_I(x) &= -x_1 \sum_{i \in I} \alpha_i(x) c_i a_i. \end{aligned}$$

The weak source singled out in the supplied proof is $\mathcal{S}_2 = \frac{1}{2}\mathbb{E}_{2,n}[(f_1 - x_1)H_I]$.

Lemma 1.1 (Exact square and signed corrections). *One has $H_I = F_I + L_I + Z_I$ and*

$$\begin{aligned} \mathcal{S}_2 &= \frac{1}{2}\|F_I\|_{2,n}^2 + \frac{1}{2}\mathbb{E}_{2,n}[F_O F_I] - \frac{1}{2}\mathbb{E}_{2,n}[x_1 F_I] \\ &\quad + \frac{1}{2}\mathbb{E}_{2,n}[(f_1 - x_1)L_I] + \frac{1}{2}\mathbb{E}_{2,n}[(f_1 - x_1)Z_I]. \end{aligned} \quad (2)$$

No sign is asserted for the four corrections.

Proof. Selector compatibility gives $F_I = \sum_I \alpha_i c_i (a_i x_1 + b_i x_2)$. Subtract this from H_I and expand $f_1 = F_I + F_O$ in its inner product with F_I . These operations are pointwise finite-sum identities and do not freeze a training gate. $\square$

Thus approximate input–output alignment is not innocuous unless its *signed contribution at the scale of the source* is controlled. Small noise in coordinates also need not give a smaller-order target term.

2 A smooth potential and exact residual dynamics

Let J_t be the selected Jacobian of the training output map from parameters to the empirical output space. The flow and residual equation are

$$\dot{\theta} = J_t^* e_t, \quad \dot{e}_t = -K_t e_t, \quad K_t = J_t J_t^* \succeq 0. \quad (3)$$

The adjoint includes empirical averaging. Matrix calculations below can instead use outputs and residuals divided by $\sqrt{N}$ and ordinary transposes.

Proposition 2.1 (The empirical statistic as a residual functional). *Define the smooth polynomial in the parameters*

$$\Phi_I(\theta) = \frac{1}{2}(e_1^T M v - r)^2 + e_2^T M v, \quad g_t = \nabla_\theta \Phi_I(\theta_t), \quad C_t = J_t g_t.$$

Then $Q_G = \dot{\Phi}_I = \langle C_t, e_t \rangle_n$. Its gradient is explicitly

$$(g_t)_{w_i} = \mathbf{1}_{i \in I} \ell(u_i^T v), \quad (g_t)_{u_i} = \mathbf{1}_{i \in I} v(w_i^T \ell).$$

Equivalently,

$$C_t(x) = \sum_{i \in I} \alpha_i(x) [(u_i^T x)(u_i^T v) \ell + (x^T v)(w_i^T \ell) w_i].$$

All neurons, including those outside I , enter e_t, J_t, K_t .

Proof. Differentiate Φ_I with respect to M to obtain ℓv^T , and then with respect to the two neuron vectors to obtain g_t . The product rule gives $\dot{\Phi}_I = \ell^T \dot{M} v = Q_G$. Substitute the flow and take the adjoint. Applying the network Jacobian to g_t gives the last formula. Equation (3) follows from the absolutely continuous ReLU chain rule, including ad preactivations that vanish on a set of positive time measure. $\square$

The positivity of K_t does not determine the sign of $\langle C_t, e_t \rangle_n$. It does, however, prevent exponential amplification in a comparison of *training residuals*. This is different from a parameter-space Grönwall comparison or a global frozen-kernel training assumption.

Theorem 2.2 (Contractive finite-window comparison). *Fix an entry time a and a horizon $H > 0$. Write $e_a = e$, $J_a = J$, $K_a = K$, $g_a = g$, $C_a = C$, and define*

$$\bar{e}(\tau) = e^{-K\tau} e, \quad \Psi(\tau) = \langle C, e^{-K\tau} e \rangle_n, \quad 0 \leq \tau \leq H.$$

For almost every τ , the exact discrepancy is

$$\begin{aligned} Q_G(a + \tau) - \Psi(\tau) &= \langle C_{a+\tau} - C, e_{a+\tau} \rangle_n \\ &\quad - \int_0^\tau \langle e^{-K(\tau-s)} C, (K_{a+s} - K) e_{a+s} \rangle_n ds. \end{aligned} \quad (4)$$

If $\|C_{a+s} - C\|_n \leq d_C(s)$ and $\|K_{a+s} - K\|_{\text{op}} \leq d_K(s)$ almost everywhere, then

$$|Q_G(a + \tau) - \Psi(\tau)| \leq \|e\|_n \left[d_C(\tau) + \int_0^\tau d_K(s) \|e^{-K(\tau-s)} C\|_n ds \right]. \quad (5)$$

In particular uniform bounds d_C, d_K give the uniform error $\rho = \|e\|_n (d_C + H d_K \|C\|_n)$. There is no exponential growth factor.

Proof. Dissipation or (3) gives $\|e_t\|_n \leq \|e\|_n$ for $t \geq a$. Variation of constants relative to K gives

$$e_{a+\tau} - e^{-K\tau} e = - \int_0^\tau e^{-K(\tau-s)} (K_{a+s} - K) e_{a+s} ds.$$

Subtract the two residual functionals and use self-adjointness of K to move its exponential onto C . This proves (4) with its signs intact. Cauchy–Schwarz proves (5). Since $K \succeq 0$, $\|e^{-Ku}\| \leq 1$ for $u \geq 0$, proving the uniform bound. The integrals only require measurable bounded Jacobians on the compact parameter path; training-gate jumps are allowed. $\square$

For a late positive witness, a lower bound on the signed expression (4) suffices. Requiring its absolute value to be small is stronger and can lose helpful drift.

Corollary 2.3 (Directional budget with signed endpoint drift). *For a fixed τ , put $b_s = J_{a+s}^* e_{a+s}$, $\Delta J_s = J_{a+s} - J_a$, and $z_s = e^{-K_a(\tau-s)} C_a$. Define*

$$P_\tau = (g_{a+\tau} - g_a)^T b_\tau + g_a^T \Delta J_\tau^* e_{a+\tau},$$

$$B_\tau = \int_0^\tau [\|\Delta J_s^* z_s\| \|b_s\| + \|J_a^* z_s\| \|\Delta J_s^* e_{a+s}\|] ds.$$

Then almost everywhere in τ ,

$$Q_G(a + \tau) \geq \Psi(\tau) + P_\tau - B_\tau. \quad (6)$$

In particular $P_\tau \geq -p$ and $B_\tau \leq B$ imply a positive witness whenever $\Psi(\tau) > p + B$. No upper bound on helpful positive endpoint drift is required. These bounds on an interval give an increasing interval for Φ_I ; a single time is not by itself such an interval.

Proof. The first term of (4) equals P_τ , since $J_t g_t - J_a g_a = J_t(g_t - g_a) + (J_t - J_a)g_a$. Moreover

$$K_{a+s} - K_a = \Delta J_s J_{a+s}^* + J_a \Delta J_s^*.$$

Thus its integrand is exactly $(\Delta J_s^* z_s)^T b_s + (J_a^* z_s)^T \Delta J_s^* e_{a+s}$. Cauchy-Schwarz bounds this above by the integrand of B_τ . This proves the lower bound, retaining the sign of P_τ . $\square$

The first integral can alternatively be bounded by

$$\sqrt{\mathcal{L}(a) - \mathcal{L}(a + \tau)} \left(\int_0^\tau \|\Delta J_s^* z_s\|^2 ds \right)^{1/2},$$

using loss dissipation. This need not be sharper than keeping the pointwise product. Neither form requires an operator-norm bound over all residual directions.

3 Spectral crossing and full-error transfer

The comparison function is calculated in parameter dimension rather than forming a $2N$ -dimensional kernel. Put $B = J^* J$, $b = J^* e$. If $B z_j = \lambda_j z_j$ is an orthonormal eigendecomposition, then

$$\Psi(\tau) = g^T e^{-B\tau} b = \sum_j c_j e^{-\lambda_j \tau}, \quad c_j = (g^T z_j)(z_j^T b), \quad \lambda_j \geq 0. \quad (7)$$

This follows from $J^* e^{-J J^* \tau} = e^{-J^* J \tau} J^*$. The coefficients are signed. Positive and negative terms can decay at different rates even though the training residual norm is nonincreasing.

Corollary 3.1 (A sufficient two-rate envelope). *Choose $0 \leq s < f$. Let A be the sum of positive c_j with $\lambda_j \leq s$, let B_- be the sum of $|c_j|$ over negative c_j with $\lambda_j \geq f$, and let D_- be the sum of $|c_j|$ over all remaining negative coefficients. Then*

$$\Psi(\tau) \geq A e^{-s\tau} - B_- e^{-f\tau} - D_-.$$

Consequently if the right side exceeds ρ on a nontrivial late interval, and $\Psi < -\rho$ on a nontrivial earlier interval, $\Phi_I(\theta_t)$ has a decreasing interval followed by an increasing interval under Theorem 2.2.

Proof. Bound retained positive terms below by $e^{-s\tau}$ times their total coefficient, the fast negative terms below by $-e^{-f\tau} B_-$, and the remaining negative terms below by $-D_-$. Discard other positive terms. Apply (5) and integrate the strict signs almost everywhere. $\square$

The full spectral sum may give a much sharper certificate than this envelope; no claim that two literal eigenmodes suffice is required.

Theorem 3.2 (Crossing localization from an entry-state response). *Suppose the comparison error is uniformly at most ρ on $[0, H]$. Assume the explicitly computed Ψ has $0 < \tau_* < H$, and for some $\kappa > 0$,*

$$\Psi(\tau) \leq -\kappa(\tau_* - \tau) \quad (\tau < \tau_*), \quad \Psi(\tau) \geq \kappa(\tau - \tau_*) \quad (\tau > \tau_*).$$

If $\rho/\kappa < \min(\tau_, H - \tau_*)$, every minimum of $\Phi_I(\theta_t)$ on $[a, a + H]$ is interior and is within ρ/κ of $a + \tau_*$. The function is strictly decreasing before this bracket and strictly increasing after it. A continuous zero of Q_G is not asserted.*

If the exact full-probe discrepancy satisfies $|\dot{E}_{\text{full}} - Q_G| \leq d$ on the same interval, replace ρ by $\rho + d$ to obtain the corresponding conclusion for E_{full} .

Proof. For $\tau < \tau_* - \rho/\kappa$, the upper bound for Q_G is strictly negative almost everywhere. For $\tau > \tau_* + \rho/\kappa$, its lower bound is strictly positive. Since $\Phi_I(\theta_t)$ is absolutely continuous, integration proves strict monotonicity on the outer intervals. Continuity and compactness give a minimum, and those intervals exclude any minimizer outside the bracket. The proof for E_{full} uses the same argument after adding d . $\square$

In contrast with assuming the actual Q_G crossing, the reference response and its slopes are calculated from one state. Reachability of that state and a sufficiently small error remain separate questions. The full discrepancy d must include inactive strong neurons, the active complement and their velocities, and the difference between Q_G and the exact reduced derivative. A forward residual bound alone does not supply d .

4 A completely explicit drift bound, including gate changes

The preceding error can be bounded from a single state without postulating a trajectory regularity constant. The following deliberately conservative bound is useful for stating exactly what has and has not been certified. Let $X_* = \max_n \|x_n\|$, $P = \|\theta_a\|$, $R_* = \sqrt{H\mathcal{L}(a)}$, and $V = \|v\|$. For each neuron define the set of possibly changing sample gates

$$\mathcal{B}_i = \{x_n : |u_i(a)^T x_n| \leq R_* \|x_n\|\}, \quad \Gamma^2 = \sum_i \|w_i(a)\|^2 \mathbb{E}_n[\|x\|^2 \mathbf{1}_{\mathcal{B}_i}].$$

Set

$$\begin{aligned} D_M &= PR_* + \frac{1}{2}R_*^2, & D_J &= \sqrt{2}X_*R_* + \Gamma, \\ D_g &= V\|\ell_a\|R_* + V^2D_M(P + R_*), & j &= \|J_a\|, \\ D_C &= D_J\|g_a\| + (j + D_J)D_g, & D_K &= (2j + D_J)D_J. \end{aligned}$$

Proposition 4.1 (Primitive finite-window error). *Throughout $[a, a + H]$, these constants bound respectively the block change, Jacobian change, potential-gradient change, C change and K change. In particular the comparison theorem holds with*

$$\rho_{\text{prim}} = \sqrt{2\mathcal{L}(a)}[D_C + HD_K\|C_a\|_n]. \quad (8)$$

No probability claim or entry-state assertion is included.

Proof. Loss dissipation and Cauchy–Schwarz imply

$$\|\theta_t - \theta_a\| \leq \int_a^t \|\dot{\theta}_s\| ds \leq \sqrt{(t-a)(\mathcal{L}(a) - \mathcal{L}(t))} \leq R_*.$$

For a fixed block, expanding $W_I^T U_I$ gives $\|\Delta M\| \leq PR_* + R_*^2/2$. A changed selector can only occur on $\mathcal{B}_i$. Split the Jacobian change into its smooth weight/activation change and its changed-selector part, attaching the latter to $w_i(a)$. The squared Hilbert–Schmidt norm of the smooth part is at most $X_*^2(2\|\Delta U\|_F^2 + \|\Delta W\|_F^2) \leq 2X_*^2R_*^2$. The selector part has Hilbert–Schmidt norm at most Γ . The operator norm is no larger; this proves D_J .

Next $\|\ell_t - \ell_a\| = |Y_t - Y_a| \leq VD_M$. The explicit formulas for g in Proposition 2.1 give $\|g_t - g_a\| \leq V\|\ell_a\|R_* + V^2D_M(P + R_*)$. Expanding $J_t g_t - J_a g_a$ and $J_t J_t^* - J_a J_a^*$ gives D_C, D_K . Apply Theorem 2.2 and $\|e_a\|_n = \sqrt{2\mathcal{L}(a)}$. $\square$

If every initial training margin exceeds $R_*\|x\|$, then $\Gamma = 0$. For bounded state norms, fixed H and fixed positive margins, writing $\epsilon_e = \|e_a\|_n$ gives $R_* = O(\epsilon_e)$, $D_J, D_g, D_C, D_K = O(\epsilon_e)$, and therefore $\rho_{\text{prim}} = O(\epsilon_e^2)$. Thus an order- ϵ_e signed spectral crossing is stable to this bound in such a small-residual regime. This scaling observation does not prove that canonical random initialization reaches either the spectral condition or the required margins.

5 Checks on existing canonical states and the remaining inequality

The reanalysis uses the saved canonical seed-0 trajectory with $h = 200$, $N = 4000$, means 3, 2, noise standard deviation 0.15, and the stored strong mask selected at time 4. The fixed probe is -85° . No training was rerun. The saved states come from Euler integration; all numbers below are retrospective floating-point diagnostics, not validated continuous-time bounds. The scripts reconstruct the original seeded dataset.

At $t = 3.60$, equation (2) has the following terms:

Term	Value
Positive square	+0.0036797850
Complement correlation	−0.0006266230
Target correlation	−0.0042214838
Input–output mismatch contribution	+0.0012836054
Gate-weighted x_1 correction	+0.0017669433
Total $\mathcal{S}_2$	+0.0018822269

The target term alone exceeds the square. The mismatch and x_1 corrections are essential to this decomposition’s sign. The normalized alignment of $f_1 - x_1$ with H_I is approximately 0.485, rather than a consequence of an unweighted weak residual norm.

The spectral reference in (7), computed separately at each listed entry time, gives these interpolated negative-to-positive crossings:

Entry a	$Q_G(a)$	Reference crossing $a + \tau_*$	$\Psi(0.5)$
2.9	−0.00724618	3.813095	−0.00136129
3.0	−0.00555350	3.768320	−0.00077795
3.2	−0.00281777	3.704165	−0.00000939
3.4	−0.00105117	3.654520	+0.00043947
3.5	−0.00045939	3.630368	+0.00059510

The supplied audit gives the actual-statistic crossing on saved states as 3.60182403, by interpolation. The frozen reference is therefore informative but not exact. Starting at $a = 3.4$, linear decomposition by the support of the *initial residual* gives at $\tau = 0.5$ a strong-cluster contribution -0.0009715782 and a weak-cluster contribution $+0.0014110470$. Both contributions evolve under the full coupled reference kernel. This is not an ablation removing the weak training cluster and is not the same partition as the instantaneous clusterwise source in the supplied audit.

What the current bound actually certifies. For the diagnostic state $a = 3.4$, $H = 0.5$, the primitive formulas give

$$\mathcal{L}(a) = 0.0041220000, \quad R_* = 0.04539824, \quad \rho_{\text{prim}} = 0.40288812.$$

The required positive-witness inequality would be

$$\rho < \Psi(0.5) = 0.0004394687840. \quad (9)$$

It is not satisfied by the primitive bound. This failure does not disprove a useful finite-window theorem. In particular it does not show that the actual signed discrepancy is large or adverse.

The sharper remaining target. For the late witness one only needs

$$\begin{aligned} & \langle C_{a+H} - C_a, e_{a+H} \rangle_n \\ & - \int_0^H \langle e^{-K_a(H-s)} C_a, (K_{a+s} - K_a) e_{a+s} \rangle_n ds > -\Psi(H) \end{aligned} \quad (10)$$

on a neighborhood of a late time (almost everywhere), together with an early negative interval. A proved absolute bound satisfying (9) is sufficient but not necessary. No estimate in the existing specialization or loss-dissipation results establishes (10) on a reached canonical window. The source square does not fill this gap either.

A smaller signed budget on the saved states. At $a = 3.4$, $H = 0.5$, retrospective evaluation of Corollary 2.3 gives

$$\begin{aligned}(g_{a+H} - g_a)^T b_H &= +0.0003228852, \\ g_a^T \Delta J_H^* e_{a+H} &= +0.0001586021, \\ P_H &= +0.0004814873, \quad B_H^{\text{sampled}} = 0.0003116022.\end{aligned}$$

The last quantity is a trapezoidal sum of the pointwise norm products, not a rigorous upper bound on the continuous-time integral. If one could prove $P_H \geq 0$ and an upper bound as small as this sampled B_H , the positive margin left by (6) would be about 0.0001278666. This shows why taking absolute values of the endpoint drift is counterproductive: the corresponding sampled absolute budget is 0.0007930895, which exceeds $\Psi(H)$. The signed Duhamel reconstruction differs from the directly evaluated discrepancy by $2.35 \cdot 10^{-7}$; this includes quadrature and Euler-trajectory error and is not a certified error bound.

This sharper calculation identifies a quantitatively plausible target but still proves no uniform inequality on the true initialized flow. Validating a fixed-seed canonical trajectory would be a distinct, computer-assisted milestone; it would not supply a high-probability iid theorem or license treating Euler snapshots as exact gradient-flow states.

Scope of the saved-state comparison. The new results isolate a route from a reached ID state to a crossing without assuming the crossing itself. They do not prove that iid isotropic initialization reaches a favorable spectral state or that its signed drift is favorable. For the comparison route, the coordinate-specific full-probe residual and velocity must also satisfy the transfer budget on the same window. The separate local certificate below instead verifies the full derivative and snapshots directly. No common noisy high-probability parameter region or canonical timing guarantee is claimed. The supplied noiseless countermodel remains compatible with every statement here: its actual statistic need not satisfy a favorable spectral certificate. The existing finite-sample theorem and cleaned appendix are unchanged.

6 Static sign certificates with all training-gate changes

This section concerns parameter neighborhoods, not their reachability from initialization. Its dataset consists of the exact binary numbers in the accompanying `canonical_witness_states.npz`. Rounding a Gaussian sample is not an iid Gaussian theorem.

Let y be a parameter center, $R > 0$, $P = \|y\|$, $\bar{P} = P + R$, and $D = PR + R^2/2$. A cluster index set C uses the full-dataset normalization $\|z\|_C^2 = N^{-1} \sum_{n \in C} \|z_n\|^2$. Put $\lambda_C \geq \|N^{-1} \sum_{n \in C} x_n x_n^T\|$. Write F_C for that cluster's contribution to the full gradient field, using the *full-network residual*; $F = F_1 + F_2$. Let F_C^0 denote its polynomial extension with the training gates fixed at y . No fixed-gate hypothesis is imposed on the actual field in the ball.

For either potential used here, set $g = \nabla \Phi$, $G_0 = \|g(y)\|$. For the empirical statistic use $\Phi = \Phi_I$ from Proposition 2.1. For the full derivative, fix the actual probe-active set at y and use $\Phi(\theta) = \frac{1}{2} \|M_{\text{act}}(\theta)\xi - \xi\|^2$. This second choice is valid throughout the ball if $\min_i |u_i(y)^T \xi| > R\|\xi\|$. In either case let v be its input vector, I its fixed index set, $P_I = \|(u_i, w_i)_{i \in I}(y)\|$, and ℓ_y its output gradient. Define

$$\begin{aligned}L_C &= \lambda_C \bar{P}^2 + \sqrt{\lambda_C} (\|e(y)\|_C + \sqrt{\lambda_C} D), \\ H_g &= \|v\|^2 (P_I + R)^2 + \|v\| (\|\ell_y\| + \|v\| (P_I R + R^2/2)).\end{aligned}$$

Here are explicit bounds for the training-gate correction. Put $q_{ni} = u_i(y)^T x_n$, $s_n = \|x_n\|$, $e_n = e(y; x_n)$, and

$$\begin{aligned}d_{ni} &= (Rs_n - |q_{ni}|)_+, & p_{ni} &= \mathbf{1}_{|q_{ni}| \leq Rs_n}, \\ b_{ni} &= |w_i(y)^T e_n| + R\|e_n\| + (\|w_i(y)\| + R)Ds_n, \\ j_i^u &= N^{-1} \sum_{n \in C} p_{ni} b_{ni} s_n, & j_i^w &= N^{-1} \sum_{n \in C} d_{ni} (\|e_n\| + Ds_n), \\ z_n &= \sum_i (\|w_i(y)\| + R) d_{ni}, & B_C &= \left[\sum_i ((j_i^u)^2 + (j_i^w)^2) \right]^{1/2} + \sqrt{\lambda_C} \bar{P} \|z\|_C.\end{aligned}$$

Every neuron is retained in these quantities.

Proposition 6.1 (A static neighborhood enclosure). *For any compatible training selectors and $\|\theta - y\| \leq R$,*

$$|g(\theta)^T F_C(\theta) - g(y)^T F_C(y)| \leq R[H_g(\|F_C(y)\| + L_C R) + G_0 L_C] + (G_0 + H_g R) B_C. \quad (11)$$

This applies to the total derivative and separately to each cluster’s signed contribution. Cluster contributions always use the same full residual.

Proof. The Jacobian norm on cluster C is at most $\sqrt{\lambda_C} \|\theta\|$: apply Cauchy–Schwarz across neuron and input/output parameter coordinates before taking the empirical average. Integration along a parameter segment gives $\|f_\theta - f_y\|_C \leq \sqrt{\lambda_C} D$ and the pointwise bound $\|f_\theta(x_n) - f_y(x_n)\| \leq D s_n$. The same bounds hold for the fixed-mask polynomial extension. Its field derivative is $-J_C^* J_C + H_{e,C}$, where the neuronwise residual Hessian has norm at most $\sqrt{\lambda_C} \|e\|_C$. Thus $\|DF_C^0\| \leq L_C$. Differentiating either potential gives $\|Dg\| \leq H_g$ on the ball. Adding and subtracting $g(y)^T F_C^0(\theta)$ proves the first term of (11).

A changed training activation differs from its fixed-mask extension by at most d_{ni} , and its selector can differ only when $p_{ni} = 1$. The pointwise residual and weight bounds give $|w_i(\theta)^T e_\theta(x_n)| \leq b_{ni}$. The direct input- and output-gradient changes are therefore bounded by j_i^u, j_i^w . The change of residual between the actual network and the fixed-mask extension has norm at most z_n . Applying the fixed-mask adjoint to this residual costs at most $\sqrt{\lambda_C} \bar{P} \|z\|_C$. Consequently $\|F_C - F_C^0\| \leq B_C$. Multiply by $\|g(\theta)\| \leq G_0 + H_g R$ to finish. Endpoint partial selectors obey the same bounds. $\square$

Verified static values. The two centers are RK4 reference states at 3.4 and 3.9, generated with step 1/5000, the original canonical dataset and the exact archived initial parameter values. The cohort is the original offline strong cohort of 71 neurons. The bounds below hold on balls of radius $3.5 \cdot 10^{-5}$. Entries in the table are outward rounded and multiplied by 10^3 .

Statistic	contribution	center at 3.4	center at 3.9
Q_G	total	$[-1.205, -0.894]$	$[0.677, 0.982]$
Q_G	strong cluster	$[-2.479, -2.198]$	$[-1.224, -0.944]$
Q_G	weak cluster	$[1.211, 1.367]$	$[1.835, 1.992]$
$\dot{E}$	total	$[-1.439, -1.156]$	$[0.449, 0.728]$
$\dot{E}$	strong cluster	$[-2.522, -2.266]$	$[-1.234, -0.978]$
$\dot{E}$	weak cluster	$[1.025, 1.168]$	$[1.622, 1.766]$

Thus weak-cluster forcing opposes the strong-cluster improvement at both states and dominates it at the later state. This is a signed decomposition of the actual field, not a retraining ablation or an assumed positive source. The full-error computation includes all probe-active neurons directly; it does not infer its sign from a small forward reduction residual.

On both balls the cohort lies in the positive strong cone: input angles are below 15° , output angles below 20° . Its ungated m_{11} is greater than 0.9322 and 0.9340, respectively. These are coarse learned-block geometry statements, not fine locking or an initialization-to-specialization theorem. The full-field speed is at most 0.385 and 0.309, respectively.

The accompanying script evaluates (11) with outward midpoint-radius bounds, dot-product rounding bounds, and interval trigonometry. All center training masks are resolved by intervals; possible changes anywhere in each ball are charged by B_C . The probe masks stay fixed because their center margins exceed $3.7575 \cdot 10^{-5}$ and $5.6601 \cdot 10^{-5}$. Exact-rational corner tests independently check 553 arithmetic enclosures. The arithmetic contract is IEEE round-to-nearest with standard dot accumulation and no unsafe reassociation; these tests are not a proof of initialized reachability.

7 An a posteriori enclosure through training-gate changes

This section states the real-arithmetic stability inequality used by the canonical certificate prototype. A successful floating-point run is not by itself a certificate; its input and rounding enclosures must also be verified.

Write $F(\theta) = J_\theta^*(x - f_\theta)$ for the selected vector field. Fix a reference state y and a radius R . Assume that along every straight parameter segment within this ball, the smooth pieces obey

$$\max_i \|\mathbb{E}_n[e_\theta(x) x^T \alpha_i(x)]\| \leq a.$$

For each neuron and sample choose an upper bound $h_{in} \geq N^{-1}\|x_n\|(w_i^T e_\theta(x_n))_+$ throughout the ball. Set $d_{in} = |u_i(y)^T x_n|/\|x_n\|$ for nonzero samples, and

$$B_i(s) = \sum_{n:d_{in} \leq s} h_{in}, \quad 0 \leq s \leq R.$$

If the center ranges along a reference subinterval, lower bounds for the margins and uniform upper bounds for the weights can replace these quantities.

Lemma 7.1 (One-sided stability with adverse gate jumps). *Suppose $B_i(s) \leq a_i + \beta s$ for $0 \leq s \leq R$, where $a_i, \beta \geq 0$. Then for $\|z\| \leq R$,*

$$\langle z, F(y+z) - F(y) \rangle \leq (a + \beta)\|z\|^2 + \|(a_i)_i\|_2\|z\|.$$

Compatible endpoint selectors are allowed. Along an absolutely continuous reference path $y(t)$ with defect $\|F(y(t)) - \dot{y}(t)\| \leq \eta$, the error $r(t) = \|\theta(t) - y(t)\|$ therefore satisfies, while $r \leq R$,

$$D^+ r \leq (a + \beta)r + \|(a_i)_i\|_2 + \eta$$

in the comparison sense (or almost everywhere away from $r = 0$).

Proof. On a smooth activation region, $DF = -J^*J + H_e$. The residual Hessian H_e is block diagonal across neurons, with each block having off-diagonal entries A_i, A_i^T , where $A_i = \mathbb{E}_n[ex^T \alpha_i]$. Thus $\lambda_{\max}(DF) \leq a$. Integrate this inequality along the segment $y + sz$, separating its finitely many training hyperplane crossings. When one gate crosses, the vector-field jump has only an input-coordinate component $\Delta F_{u_i} = N^{-1}\Delta \alpha_i(w_i^T e_x)x$; the output-coordinate component is continuous because the preactivation is zero there. Its pairing with z is at most $N^{-1}(w_i^T e_x)_+ |z_{u_i}^T x| \leq h_{in}\|z_{u_i}\|$. Only samples with $d_{in} \leq \|z_{u_i}\|$ can cross. The sum of jumps is therefore bounded by

$$\sum_i \|z_{u_i}\| B_i(\|z_{u_i}\|) \leq \|(a_i)_i\|_2\|z\| + \beta\|z\|^2.$$

Simultaneous crossings add; endpoint partial selectors satisfy the same upper bound by taking partial jumps. Segments that lie in a hyperplane contribute no conormal pairing for that gate. This proves the first claim. Take the inner product of the error equation with $\theta - y$ and divide by r when it is positive. Standard norm regularization gives the comparison at zero. $\square$

For a finite set of distances, permissible intercepts are explicit:

$$a_i(\beta) = \max \left\{ 0, \max_{0 \leq s \leq R} (B_i(s) - \beta s) \right\}.$$

The inner maximum only needs the sorted sample distances, including zero. There is no density assumption or asymptotic substitution for these counts.

Corollary 7.2 (A verified step criterion). *On a reference interval of length h , let the preceding bounds be uniform on its radius- R tube. Put $k = a + \beta$ and $\omega = \|(a_i)_i\|_2 + \eta$. If the initial error is at most $r_0 < R$ and*

$$e^{kh} r_0 + \omega \frac{e^{kh} - 1}{k} < R,$$

then the flow remains in the tube on the whole interval and its terminal error is bounded by the left side. At $k = 0$ the quotient is interpreted as h . Successive verified steps give an enclosure from the initial state.

Proof. Apply the scalar integrating-factor inequality until the first exit. Its upper bound is nondecreasing in time and strictly smaller than R at the terminal time, excluding a first exit. Iterate using the terminal upper bound as the next initial radius. $\square$

The absolute reference defect must also include any gates crossed by the reference path. The favorable sign used in Lemma 7.1 applies to *stability*, not automatically to the defect of a numerical interpolant. Using only a smooth-region defect bound would omit this term and would not constitute a certificate.

Proposition 7.3 (Sensitivity update at one transverse training-gate event). *Suppose exactly one training gate (i, n) switches at $u_i^T x_n = 0$, all other gates are locally fixed, and the two smooth one-sided vector fields F^-, F^+ are transverse in the same direction. Let ν be the full parameter vector with x_n in the u_i coordinates and zero elsewhere. The linearized, time-aligned perturbation update at this event is*

$$S = I + \frac{(F^+ - F^-)\nu^T}{\nu^T F^-} = I + \frac{\Delta\alpha_i(w_i^T e_{x_n})}{N\nu^T F^-}\nu\nu^T.$$

Its nontrivial eigenvalue is $\nu^T F^+/\nu^T F^- > 0$ and

$$\|S\|_2 = \max\{1, \nu^T F^+/\nu^T F^-\}.$$

In particular adverse expansion corresponds to $w_i^T e_{x_n} > 0$; negative conormal residuals do not increase the Euclidean operator norm at the event. This is a local linearized identity, not a nonlinear tube bound.

Proof. For a perturbation z^- at the nominal event time, first-order adjustment of the hitting time is $\delta t = -\nu^T z^-/(\nu^T F^-)$. Transport to the perturbed event with F^- and back to the common nominal time with F^+ : $z^+ = z^- + (F^- - F^+)\delta t$. This gives the first formula. The network output and training residual are continuous at the gate, and its activation is zero, so only the input-gradient term jumps. Its jump is $\Delta\alpha_i(w_i^T e_{x_n})\nu/N$, proving the second formula. This is a symmetric rank-one update: it is identity on $\nu^\perp$ and has eigenvalue $(\nu^T F^+)/(\nu^T F^-)$ along ν . Same-direction transversality makes this eigenvalue positive. Finally $\Delta\alpha_i$ has the sign of $\nu^T F^-$, which proves the last claim. $\square$

An event-resolved certificate must verify transversality, enclose the event state and time, propagate nonlinear errors, and control floating-point rounding. Merely multiplying sampled sensitivity ratios or observing that sampled velocities have the same sign does not complete these steps.

7.1 Cubic-reference defect used in the local validator

Let $y(s)$ be the cubic Hermite interpolant of two stored parameter states and two stored velocities, over a step of exact length $1/5000$. The stored velocities are merely coefficients of the reference polynomial; they need not be exact field values. Put $a = 1/10000$ and center the step at $s = 0$. Fix the actual training masks at $y(0)$ and denote that polynomial field by F^0 . The validator encloses

$$\begin{aligned} d_0 &= \|F^0(y) - y'\|, \\ d_1 &= \|DF^0(y)y' - y''\|, \\ d_2 &= \|D^2F^0(y)[y', y'] + DF^0(y)y'' - y'''\| \end{aligned}$$

at the midpoint. Suppose P_*, V_*, A_*, T_* bound respectively $\|y\|, \|y'\|, \|y''\|, \|y'''\|$ throughout the step, and L_* bounds $\|DF^0\|$. For the covariance bound λ ,

$$\|D^2F^0\| \leq 3\lambda P_*, \quad \|D^3F^0\| \leq 3\lambda.$$

Indeed the fixed-mask output is quadratic with second derivative norm at most $\sqrt{\lambda}$, while $\|J\| \leq \sqrt{\lambda}P_*$; differentiate $F^0 = J^*e$ twice and three times. Taylor's integral remainder therefore gives

$$\eta_{\text{sm}} = d_0 + ad_1 + \frac{1}{2}a^2d_2 + \frac{1}{6}a^3(3\lambda V_*^3 + 9\lambda P_*V_*A_* + L_*T_*)$$

as an upper bound for the fixed-mask defect on the whole step. The third derivative of the defect uses $y'''' = 0$.

The true defect additionally includes η_{gate} , the difference between the actual field and F^0 along the reference polynomial. It is bounded by the direct field argument in Proposition 6.1, using neuronwise input and output excursions

$$d_i^u = a\|y'_{u_i}(0)\| + \frac{1}{2}a^2\|y''_{u_i}(0)\| + \frac{1}{6}a^3\|y'''_{u_i}\|,$$

and the analogous d_i^w . The corresponding whole-parameter excursion bounds residual changes. Thus $\eta = \eta_{\text{sm}} + \eta_{\text{gate}}$ is uniform on the step, including all reference-path gate changes.

For the one-sided rate on a tube, let A_i^0 be its midpoint residual matrices and D_* the pointwise output-change coefficient from the midpoint throughout the tube. If p_{in} flags every potentially changed gate there, a valid smooth-piece bound is

$$a_* = \max_i \|A_i^0\| + \lambda D_* + \max_i N^{-1} \sum_n p_{in} \|e_y(x_n)\| \|x_n\|.$$

For gate jumps the validator bounds the positive conormal residual throughout the same tube, sorts lower bounds on its reference-path gate distances, and sums upward-rounded weights. With $\beta = 1$, the intercept $\max_s (B_i(s) - s)_+$ only requires those finite sorted distances. All tube radii are explicit inputs, rather than inferred from successful numerical integration. Corollary 7.2 is checked at every step. Its strict inequality is what validates the reference, independently of the integration method originally used to choose its coefficients.

8 A local noisy many-neuron reversal certificate

The following result is a conditional entry theorem. Its hypothesis specifies a neighborhood of an ID-trained state; it does not assume an OOD derivative sign or a crossing. The local dynamics and the OOD conclusions are certified below. Initialization-to-entry is a separate unresolved inequality.

Let X be the supplied canonical binary dataset, $h = 200$, and $\hat{\theta}_a$ the supplied numerical center indexed by $a = 3.4$. The loss uses all $N = 4000$ samples. The fixed cohort I is the original 71-neuron offline strong cohort. Let

$$\xi_\alpha = (\cos \alpha, \sin \alpha), \quad \mathcal{I} = \{\alpha : |\alpha + 17\pi/36| \leq 10^{-6}\}.$$

The sector has angular width $2 \cdot 10^{-6}$ radians.

Theorem 8.1 (Reversal from a certified learned-state neighborhood). *Assume a compatible gradient-flow solution satisfies*

$$\|\theta(3.4) - \hat{\theta}_{3.4}\|_2 \leq 10^{-6}. \quad (12)$$

Then, uniformly for $\alpha \in \mathcal{I}$,

$$\begin{aligned} E(\xi_\alpha, 3.4) - E(\xi_\alpha, 3.7) &> 1.2 \cdot 10^{-4}, \\ E(\xi_\alpha, 3.9) - E(\xi_\alpha, 3.7) &> 4 \cdot 10^{-5}. \end{aligned}$$

In particular the full-network error attains an interior minimum on $[3.4, 3.9]$. At entry the fixed cohort has $m_{11} > 0.9322$, input angles below 15° , and output angles below 20° . For the central probe, $Q_G < 0$ at entry and $Q_G > 0$ at 3.9 for every compatible training selector; the full-error rate has the same strict signs. More quantitatively, for every probe in $\mathcal{I}$, $\dot{E} < -0.00115$ almost everywhere on $[3.4, 3.40001]$, and $\dot{E} > 0.000444$ almost everywhere on $[3.9, 3.90001]$. No continuous or unique zero of Q_G is asserted.

Proof and computational obligations. The local enclosure uses 2500 cubic-reference steps of length $1/5000$. At each step, outward bounds for the uniform reference defect, smooth one-sided rate, and adverse gate jumps are inserted into Corollary 7.2. The input radius is 10^{-6} , rounded upward for the scalar replay. The certified radii at 3.7 and 3.9 are below $1.199 \cdot 10^{-5}$ and $2.217 \cdot 10^{-5}$, respectively. Every strict tube-invariance inequality must pass; the complete list is `LOCAL_FLOW_ENCLOSURE.csv`, with its terminal report in `LOCAL_FLOW_CERTIFICATE.json`. These computations certify that the flow is within $3.5 \cdot 10^{-5}$ of the reference snapshots at 3.7 and 3.9. Section 6 proves the static rate and geometry bounds used here. The computational proof is conditional on the arithmetic contract stated there; source code and exact binary inputs are supplied.

For completeness, the independent snapshot verification gives

$$\begin{aligned} E(\xi_0, 3.4) &\in [0.48314414, 0.48315477], \\ E(\xi_0, 3.7) &\in [0.48298853, 0.48299930], \\ E(\xi_0, 3.9) &\in [0.48306002, 0.48307086], \quad \xi_0 = \xi_{-17\pi/36}. \end{aligned}$$

The displayed intervals are enlarged from the full-precision certified values. They follow by Taylor's theorem for the full probe-error potential, with error bound $\|g(y)\|R + H_g R^2/2$; all probe gates are constant on the respective static balls. The first and second central gaps exceed 0.00014484 and 0.00006072.

For any parameter vector in these balls, $\sum_i \|w_i\| \|u_i\| \leq \|\theta\|^2/2$. Thus the full error on the unit circle is Lipschitz in the probe with constant at most $(1 + \|\theta\|^2/2)^2 < 9.084$. Moving the probe by at most 10^{-6} in angle changes a difference of two errors by less than $1.8168 \cdot 10^{-5}$. This proves the two claimed uniform gaps. Continuity gives an interior minimum. The mass, cone, and central derivative conclusions are the corresponding static certificates. The terminal enclosure is smaller than $3 \cdot 10^{-5}$. The speed bounds 0.385 and 0.309 ensure that extending either endpoint forward by 10^{-5} stays in its radius- $3.5 \cdot 10^{-5}$ static ball, by a first-exit argument. Across

the probe sector the masks remain unchanged: the additional score movement is at most $2.01 \cdot 10^{-6}$, below the unused central probe margin. On either ball

$$|\dot{E}(\xi) - \dot{E}(\xi_0)| \leq 2(1 + P^2/2)P\|F\| \|\xi - \xi_0\| < 5 \cdot 10^{-6}, \quad P < 2.01.$$

Subtract this cost from the static rate margins to obtain the asserted uniform decreasing and increasing intervals. This also extends the compatible flow through the short terminal interval. $\square$

What the signed mechanism says. At both endpoints the strong-cluster contribution to the full OOD rate is negative and the weak-cluster contribution is positive. The latter dominates at 3.9. Moreover its late lower bound exceeds its early upper bound, while the strong-cluster contribution becomes less negative, by strict certified endpoint margins. These are contributions to the *same* gradient field and full-network residual, not results of removing a cluster and retraining. They explain the local reversal by signed competition between the two training clusters. They do not prove that specialization alone is sufficient or that it causes reversal for every specialized state.

The remaining initialized obligation. The certificate starts from the small neighborhood in (12). It does not prove that the canonical initialized flow reaches it. In particular, restarting a numerical or exact flow at $\hat{\theta}_{3.4}$ is not the same as certifying the path from the archived seed. An initialized canonical theorem still requires (12), or another proved entry enclosure for which the local bounds close. The high-probability iid theorem requires an additional probability argument and is not implied by this fixed-dataset result.